MCTR903, Winter Term 10-11

Lecture 3 – Monday, October 11, 2010

# System Modeling II

These slides are based on materials from the following books:

- W. Bolton. *Mechatronics: A Multidisciplinary Approach*. 4th Edition, Pearson, 2008.
- J. Van de *Vegte. Feedback Control Systems. Prentice-Hall, 1986.*
- R. Dorf and R. Bishop. *Modern Control Systems*. 9th Ed. Prentice Hall, 2001.
- J. Doyle, B. Francis, A. Tannenbaum. *Feedback Control Theory*. Macmillan Publishing Co., 1990.



## Objectives

When you have finished this lecture you should be able to:

- Devise models for feedback systems.

# Outline

- Feedback Systems
- Feedback using Extrinsic Elements
- Feedback using Intrinsic Elements
- Case Study-1: Disk Drive
- Case Study-2: Electric Traction Motor
- Summary



# Outline

- **<u>Feedback Systems</u>**
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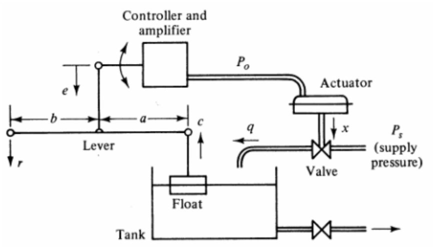
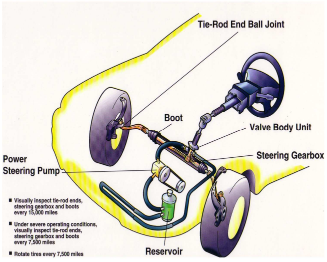
# Feedback Systems

One classification of control system is the following:

- **Process control or regulator systems:** the controlled variable, or output, must be held as close as possible to a usually constant desired value, or input, despite any disturbance.
- **Servomechanisms:** The input varies and the output must be made to follow it as closely as possible.




Water Level Control (Process Control)




Power Steering (servomechanism)

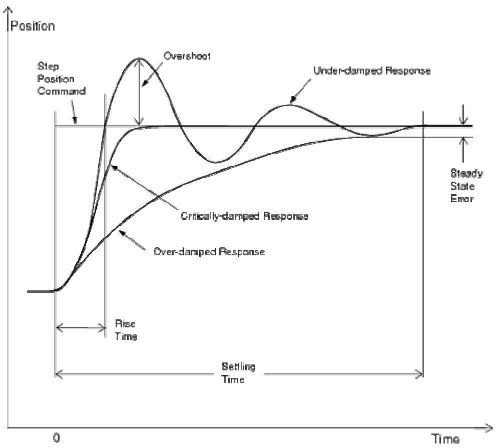


# Feedback Systems

Feedback describes the situation when **output from an event or process in the past** will influence an occurrence or occurrences of the same event/process in the **present or future**.

**Motivations for feedback:**

- Reducing the effects of parameter variations;
- Reducing the effects of disturbance inputs;
- Improving transient response characteristics
- Reducing steady-state errors.



## Feedback Systems

When feedback acts in response to an event/phenomenon, it can influence the input signal in one of two ways:

- **Positive feedback:** tends to increase the event that caused it, such as in a nuclear chain-reaction. It is also known as a **self-reinforcing loop**. An event influenced by positive feedback can increase or decrease its output/activation until it hits a **limiting constraint**. Such a constraint may be destructive, as in thermal runaway or a nuclear chain reaction. Self-reinforcing loops can be a **smaller part of a larger balancing loop**, especially in biological systems such as regulatory circuits.

Peter M. Senge (1990). The Fifth Discipline: The Art and Practice of the Learning Organization. New York: Doubleday. pp. 424. ISBN 0-385-260-946.



## Feedback Systems

- **Negative feedback:** tends to reduce the input signal that caused it, is also known as a **self-correcting or balancing loop**. Such loops tend to be **goal-seeking**, as in a thermostat, which compares actual temperature with desired temperature and seeks to reduce the difference. Balancing loops are sometimes prone to hunting: an **oscillation** caused by an **excessive or delayed negative feedback** signal, resulting in **over-correction**, wherein the signal becomes a positive feedback.

Negative feedback $\Rightarrow$ stability

Positive feedback $\Rightarrow$ unstable and explosive situations

Peter M. Senge (1990). The Fifth Discipline: The Art and Practice of the Learning Organization. New York: Doubleday. pp. 424. ISBN 0-385-260-946.

## Feedback Systems

Feedback can be classified into:

- **Feedback using extrinsic elements:** feedback is realized by **separate physical elements** and the structure of the block diagram is easy to perceive.
- **Feedback using intrinsic elements:** feedback is generated by the use of signals or physical elements which are **an intrinsic part of the system**. The precise nature of the feedback, or even its presence, may then be far from obvious. In such cases block diagrams can be derived directly from the system equations, and serve an important function in clarifying system behavior.



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- **<u>Feedback using Extrinsic Elements</u>**
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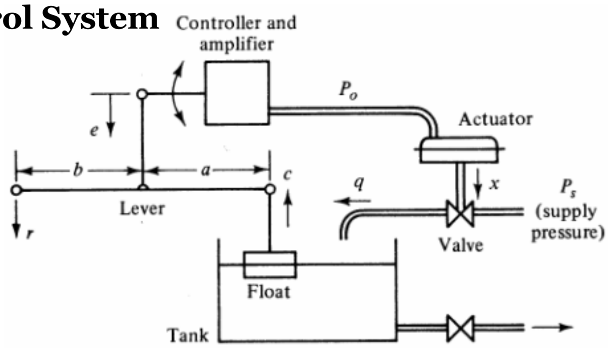
# Feedback using Extrinsic Elements

- **Water Level Control System**



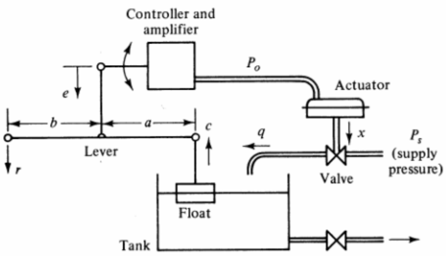

This system operates as a process control or regulator system; that is, the desired level is usually constant and the actual level must be held near it despite disturbances. The model should therefore allow for water supply pressure variations, probably the main disturbance.



# Feedback using Extrinsic Elements

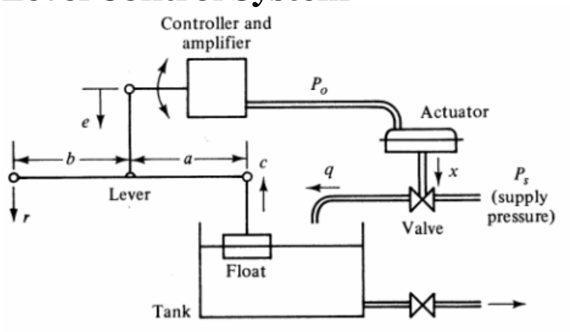
- **Water Level Control System**

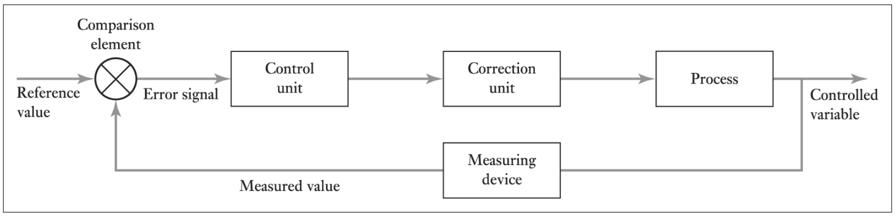
The level c is measured by means of a float and a level is used as a summing junction to determine a measure e of the error with the desired level r. From the mechanical input, the controller and amplifier sets a pneumatic output pressure Po of sufficient power to operate the pneumatic actuator which adjusts the control valve opening x to control inflow q of the tank.



# Feedback using Extrinsic Elements

- **Water Level Control System**

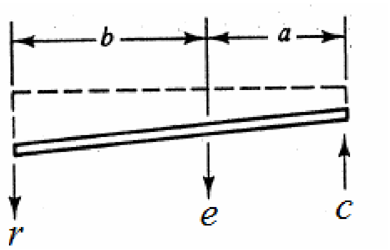





# Feedback using Extrinsic Elements

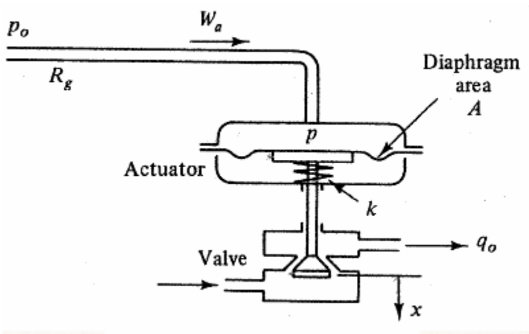
- **Water Level Control System: Mechanical Lever**




$$e = \frac{a}{a+b} r - \frac{b}{a+b} c$$

# Feedback using Extrinsic Elements

- **Water Level Control System: Pneumatic Valve Actuator**

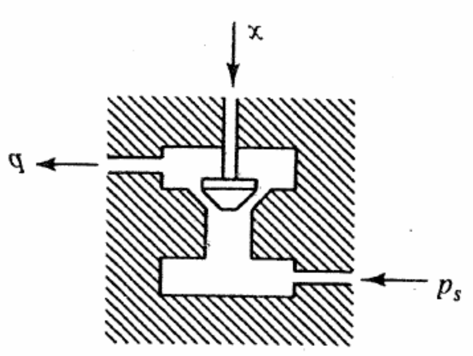



$$\frac{x}{P_o(s)} = \frac{A/k}{R_g C_g s + 1}$$



# Feedback using Extrinsic Elements

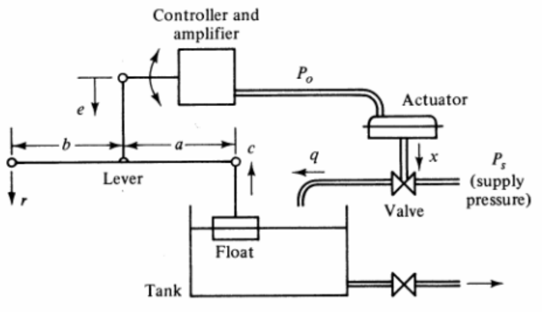
- **Water Level Control System: Control Valve, with allowance for disturbance in supply pressure $P_s$**




$$q = K_x x + K_p p_s$$

# Feedback using Extrinsic Elements

- **Water Level Control System: Tank and outflow valve**

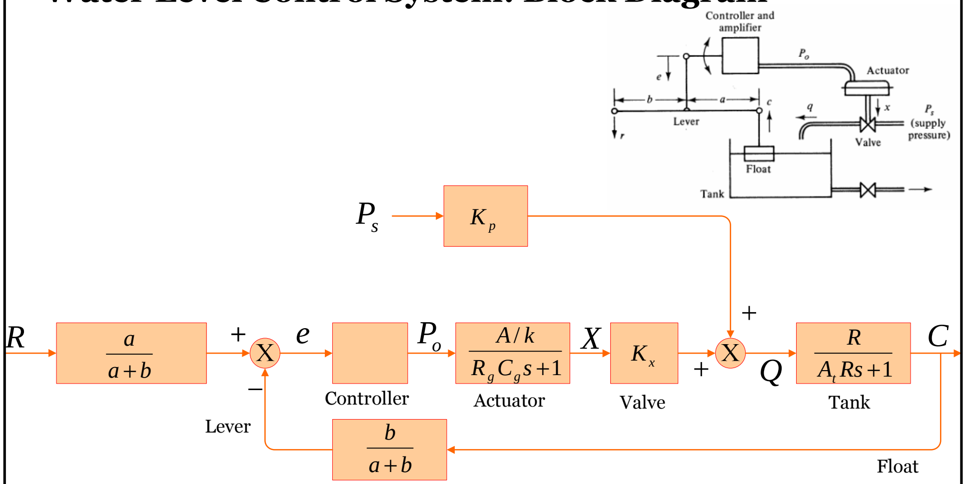



$$Q(s) = \frac{R}{(A_t Rs + 1)} C(s)$$



# Feedback using Extrinsic Elements

- **Water Level Control System: Block Diagram**

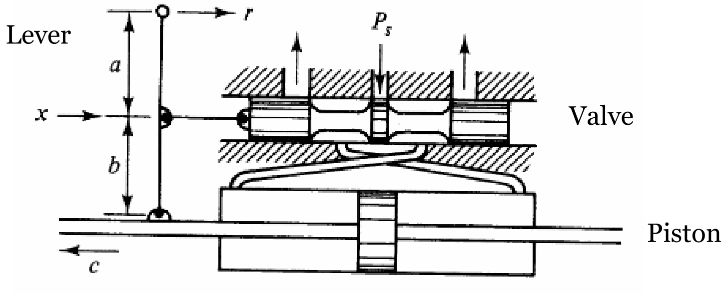



The controller and amplifier are off-the-shelf instruments.

# Feedback using Extrinsic Elements

- **Hydraulic Servo with Mechanical Feedback**

  This system consists of a **spool-valve-controlled actuator** and a mechanical level that acts as feedback element because it causes piston motion c to affect valve position x. If say, input r is moved to the right initially, the valve moves right, causing the piston to move left until the valve is again centered.



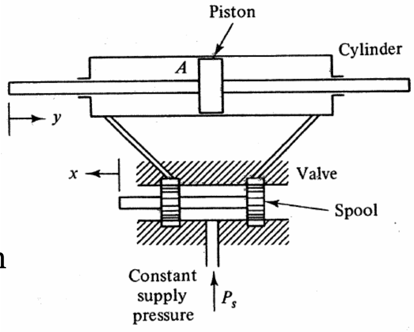



# Feedback using Extrinsic Elements

- **Hydraulic Servo: Spool-valve-controlled actuator**

  Spool-valve-controlled actuator is used extensively as fluid power control element. For x=0 the valve spool is centered, and the lands on this spool exactly block the ports of fluid lines to the ends of the cylinder, so that the piston is stopped.

  If the valve spool is moved slightly to the left, the ports are partially unblocked. The left side of the cylinder is now connected to the supply and the right side to a low-pressure reservoir. Thus the piston can move to the right.

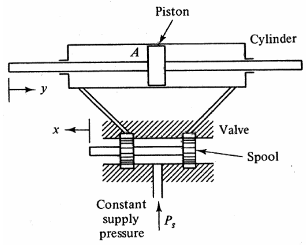


# Feedback using Extrinsic Elements

- **Hydraulic Servo: Spool-valve-controlled actuator**

Consider first the simplest possible model, in which the load connected to the piston is very small and pressure variations are negligible. Oil compressibility can then be ignored, and if the effective area A on both sides if the piston is the same, the flow rate q through both valve ports us also the same and can be modeled as

$q = K_v x$ or $Q(s) = K_v X(s)$




The change $A\dot{y}$ of volume on each side of the piston per second must equal q:

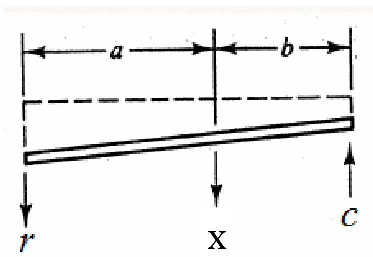
$q = A\dot{y}$ or $Q = AsY$

$$X \rightarrow \boxed{K_v} \rightarrow \boxed{1/As} \rightarrow Y$$



# Feedback using Extrinsic Elements

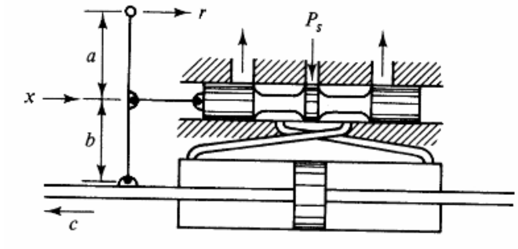
- **Hydraulic Servo: Mechanical Lever**

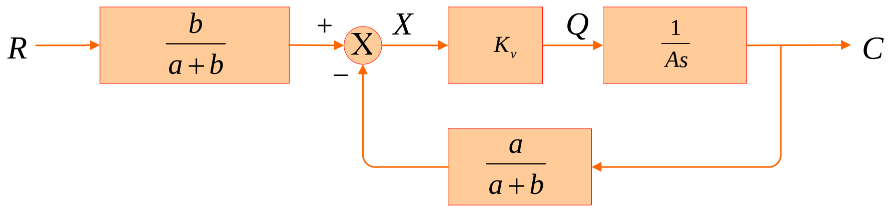



$$x = \frac{b}{a+b} r - \frac{a}{a+b} c$$

# Feedback using Extrinsic Elements

- **Hydraulic Servo: Block diagram**



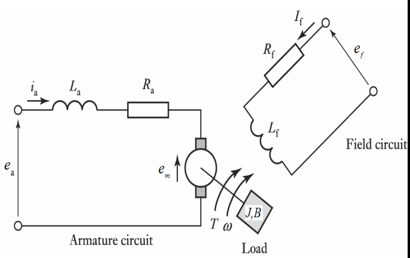



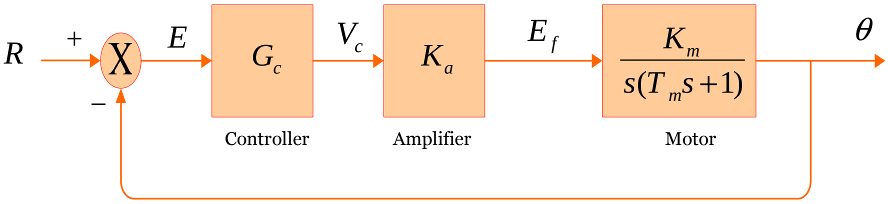



# Feedback using Extrinsic Elements

- **Simple Motor Position Servo**

The motor and its load are represented by transfer function for a field-controlled DC motor where $T_m$=J/B is the torque constant, J is the inertia and B is the damping constant.



## Feedback using Extrinsic Elements

- **Simple Motor Position Servo**

A potentiometer could represent the shaft position by a voltage and an op amp could serve as a summing junction to determine the error voltage E with voltage R representing the desired position. Bridge circuits could also be used.

The controller $G_c$ may be simple amplifier and generates a low-power output $V_c$. Its power is raised in a power amplifier of which the output is applied to the motor.



## Feedback using Extrinsic Elements

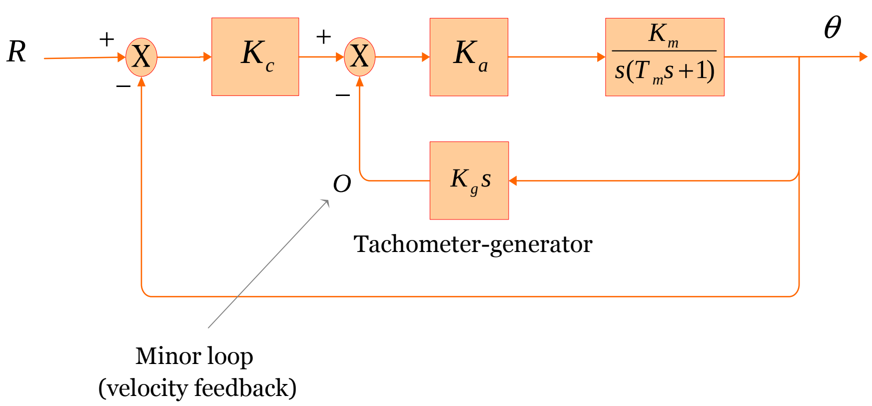
- **Servo with Velocity Feedback**

In speed or position control servomechanisms, design for satisfactory performance is often complicated by a **lack of adequate inherent damping in motor and load**. The difficulty of positioning a large inertia J rapidly without severe overshoot in response to a step input if the damping constant B is small can be appreciated intuitively. One possible solution is to install a **mechanical damper** on the motor shaft. However, a better and more elegant solution is possible by the **use of feedback**. A damping torque proportional to shaft speed $\dot{\theta}$ and in the opposite direction. Such a torque can also be generated by mounting a **small tachometer-generator** on the motor shaft to obtain a signal proportional to speed.

# Feedback using Extrinsic Elements

- **Servo with Velocity Feedback**

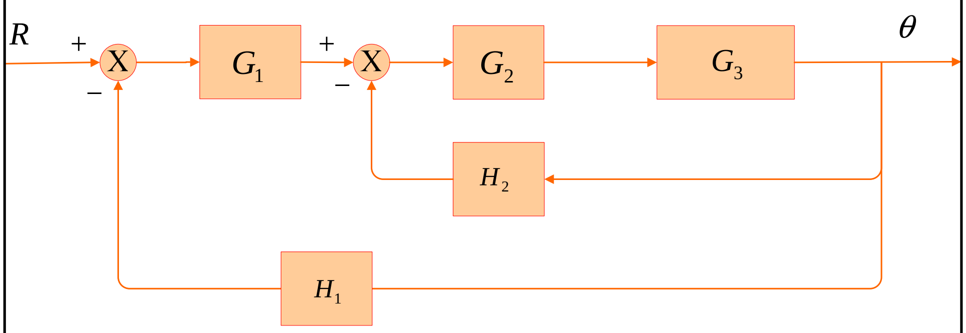
$$o = K_g \dot{\theta} \qquad \text{or} \qquad O(s) = K_g s \theta(s)$$






# Feedback using Extrinsic Elements

- **Servo with Velocity Feedback**




$$\frac{\theta}{R} = \frac{G_1 G_2 G_3}{1 + G_2 G_3 H_2 + G_1 G_2 G_3 H_1}$$

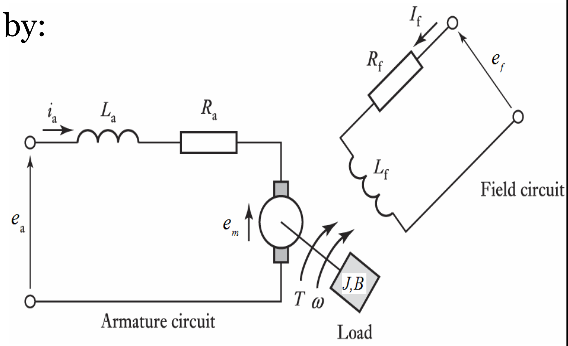
# Feedback using Extrinsic Elements

- **Motor Position Servo with Load Disturbance Torques**

  To improve the model of a field-controlled dc motor position servo by making allowance for load disturbance torques $T_l$ acting on the motor shaft, it is necessary to return to the motor equations. Field voltage $E_f$ and motor developed torque T are related to field current $I_f$ by:

$$E_f = (R_f + L_f s) I_f$$

$$T = K_t I_f$$





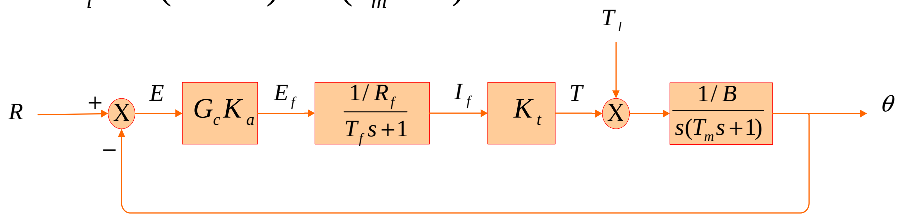

# Feedback using Extrinsic Elements

- **Motor Position Servo with Load Disturbance Torques**

  If $T_l$ is taken positive in a direction opposite that of T, a net torque $(T-T_l)$ is available to accelerate motor and load inertia J and overcome their damping B:

$$T - T_l = J\ddot{\theta} + B\dot{\theta} \qquad \text{or} \qquad T(s) - T_l(s) = s(Js+B)\theta(s)$$

$$\frac{\theta}{T - T_l} = \frac{1}{s(Js+B)} = \frac{1/B}{s(T_m s+1)} \;, \quad T_m = J/B$$



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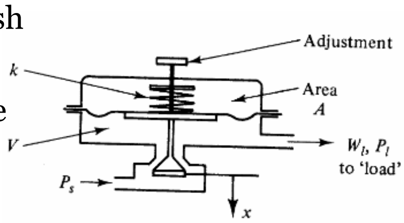
# Feedback using Intrinsic Elements

- **Pneumatic Pressure Regulator**

  The objective of this pressure regulator is to keep the pressure $P_l$ to the load serviced by the controlled constant, equal to a value set by manual adjustment, despite variations if the flow $W_l$ required by the load. Physically, the action is that a reduction $P_l$ reduces the pressure against the bottom of the diaphragm.

  This permits the spring force to push it downward to increase valve opening x, and hence increase valve flow from a supply with constant pressure $P_s$. This increase serves to raise $P_l$ back toward the set value.



# Feedback using Intrinsic Elements

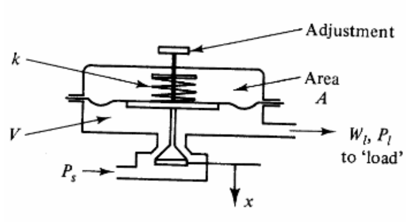
- **Pneumatic Pressure Regulator**

A linearized model is sued, implying that the variables are deviations from operating point values. Hence the constant supply $P_s$ will not appear. For the weight flow $W_v$ through the valve the following linearized model is used:

$$W_v = K_x x - K_p P_l$$

$$W_v(s) = K_x X(s) - K_p P_l(s)$$

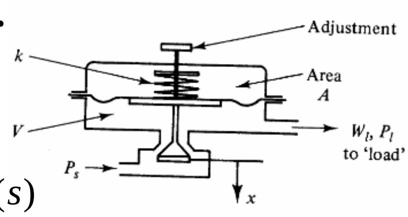
The net flow entering the volume below the diaphragm, equal to V at the operating point, is ($W_v$-$W_l$)






# Feedback using Intrinsic Elements

- **Pneumatic Pressure Regulator**




This flow must equal:

$$W_v - W_l = C_g \dot{P}_l - \rho A \dot{x}$$

$$W_v(s) - W_l(s) = C_g s P_l(s) - \rho A s X(s)$$

Here $C_g$=V/(nRT) is the capacitance of V and $C_g \dot{P}_l$ the compressibility flow.

The term $\rho A \dot{x}$ is the **flow rate** corresponding to the change of V. The minus sign arises because x is positive in the direction of decreasing volume below diaphragm.

## Feedback using Intrinsic Elements

- **Pneumatic Pressure Regulator**

  The force equilibrium on the moving parts, modeled as a spring-mass-damper system k,m,b, gives the equation

  $$m\ddot{x} + b\dot{x} + kx = -AP_l$$

  $$(ms^2 + bs + k)X(s) = -AP_l(s)$$

  This model is approximate because it neglects the flow forces on the poppet. The minus sign is needed because the pressure force $AP_l$ on the bottom of the diaphragm acts in the direction of negative x.



## Feedback using Intrinsic Elements

- **Pneumatic Pressure Regulator**

  To combine these equations into a block diagram, it is noted first that:

  ➢The output is the controlled variable $P_l$

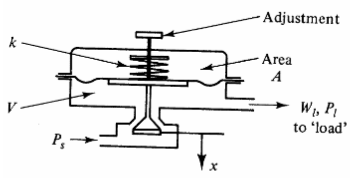
  ➢The input is the disturbance $W_l$, the unknown flow to the load.

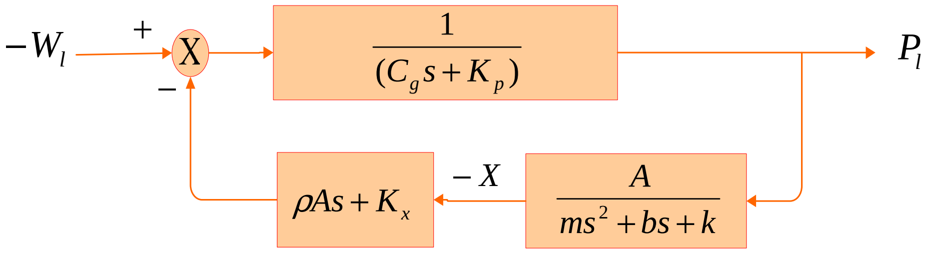
  Thus following convention, it is desirable to show $W_l$ at the left and Pl at the right in the diagram. The feedback should then show how changes of $P_l$ are used o make valve flow $W_v$ ‘follow” $W_l$. Pressure $P_l$ determines x but x and $P_l$ together determine $W_v$, and for model simplicity it appears preferable to eliminate $W_v$ between the previous equations.

# Feedback using Intrinsic Elements

- **Pneumatic Pressure Regulator**

$$K_x X - K_p P_l - W_l = C_g s P_l - \rho A s X \quad \text{or}$$
$$(C_g s + K_p) P_l = -W_l + (\rho A s + K_x) X$$



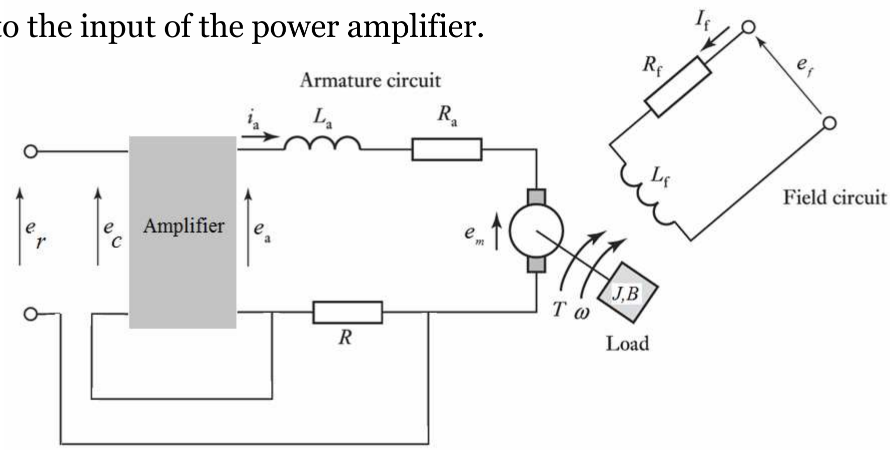






# Feedback using Intrinsic Elements

- **Motor with IR-drop Compensation**

A resistance R is added in the armature loop of an armature-controlled dc motor. The voltage across this resistor is fed back to the input of the power amplifier.



# Feedback using Intrinsic Elements

- **Motor with IR-drop Compensation**

By replacing $R_a$ by $R_a$+R

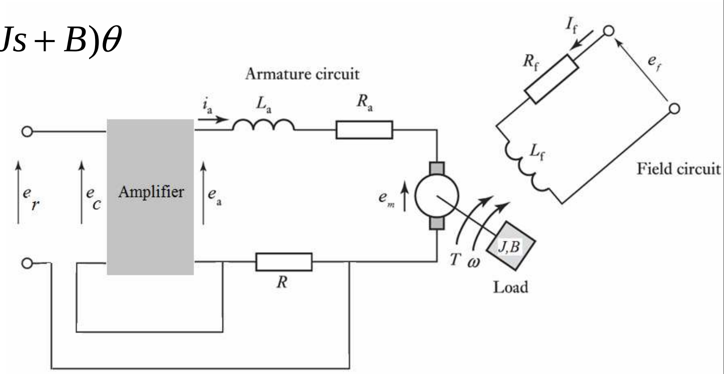
$$E_a = (R_a + R + L_a s) I_a + E_m$$

$$E_m = K_e s\theta$$

$$T = K_t I_a = s(Js + B)\theta$$

$$E_c = E_r - I_a R$$

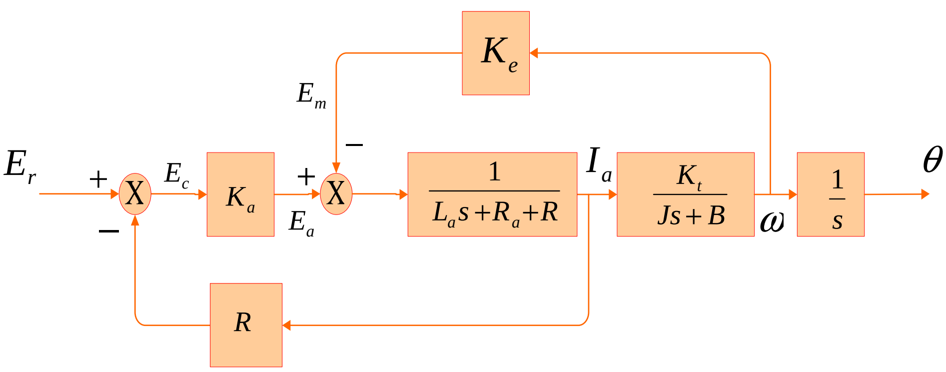
$$E_a = K_a E_c$$





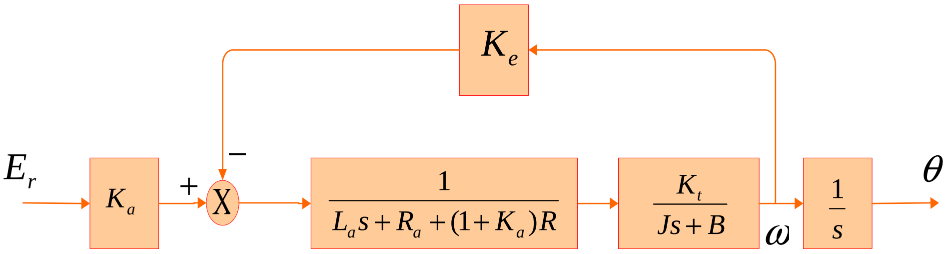

# Feedback using Intrinsic Elements

- **Motor with IR-drop Compensation**



## Feedback using Intrinsic Elements

- **Motor with IR-drop Compensation**



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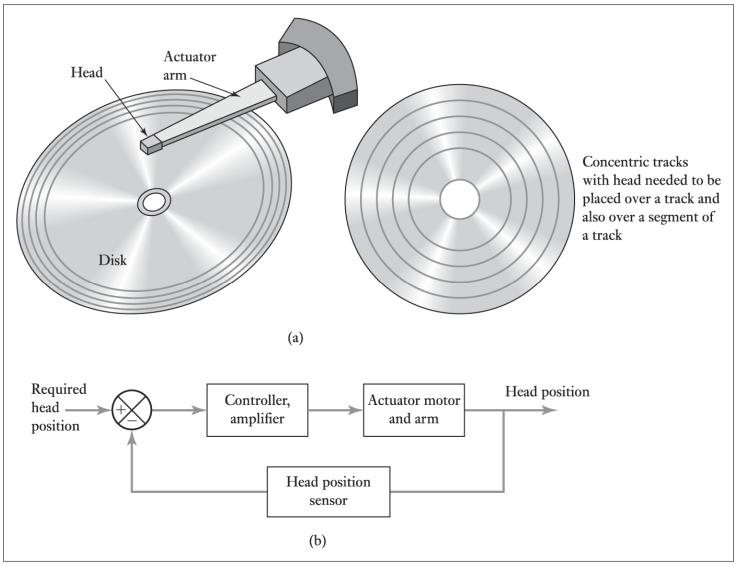
# Case Study-1: Disk drive

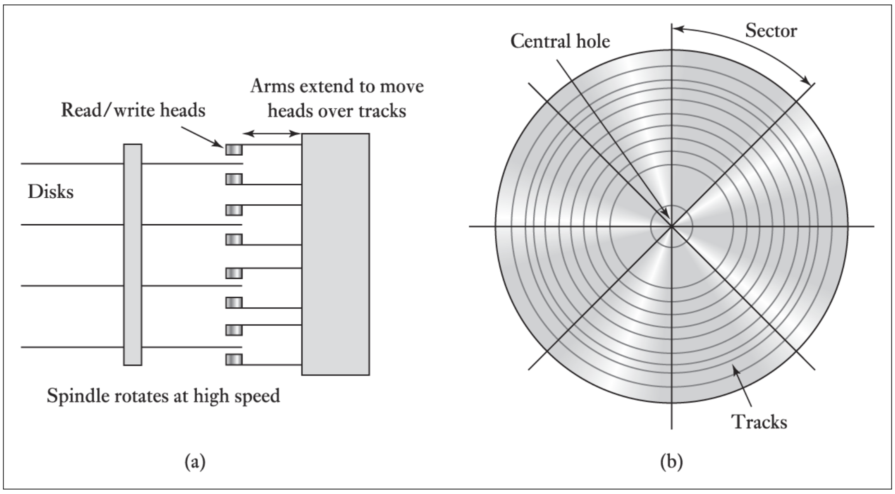



Disk drive: (a) basic form, (b) basic closed-loop control system for positioning of the read/write head



# Case Study-1: Disk drive

- **About Disk Drives**




Hard disk: (a) arrangement of disks, (b) tracks and sectors

# Case Study-1: Disk drive

- **About Disk Drives**
  - Disk drive consists of a disk coated with a metal layer which can be magnetized.
  - The gap between the write/read head and the disk surface is very small, about 0.1µm.
  - The data is stored in the metal layer as a sequence of bit cell.

```
index marker,
Sector 0 header, sector 0 data, sector 0 trailer,
Sector 1 header, sector 1 data, sector 1 trailer,
Sector 2 header, sector 2 data, sector 2 trailer,
Etc.
```

The index maker contains the track number with the sector header identifying the sector. The sector trailer contains information, e.g. a cyclic redundancy check, which can be used to check that a sector was read correctly.

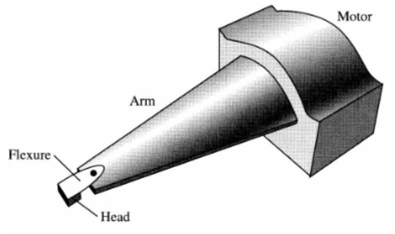


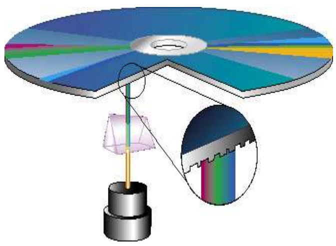
# Case Study-1: Disk drive

- **About Disk Drives**
  - The disk is rotated by a motor at typically 3600, 5400 or 7000 rpm and an actuator arm has to be positioned so that the relevant concentric track and relevant part of the track come under the read/write head at the end of that arm.
  - The head is controlled by a closed-loop system in order to position it.
  - Control information is written onto a disk during the formatting process, this enabling each track and sector of track to be identified.
  - The control process then involves the head using this information to go to the required part of the disk

# Case Study-1: Disk drive

- **About Disk Drives**
  - ➢ The actuator is a form of field-controlled d.c. motor.
  - ➢ The head element reads the magnetic field on the disk and provides a feedback signal to the control amplifier.

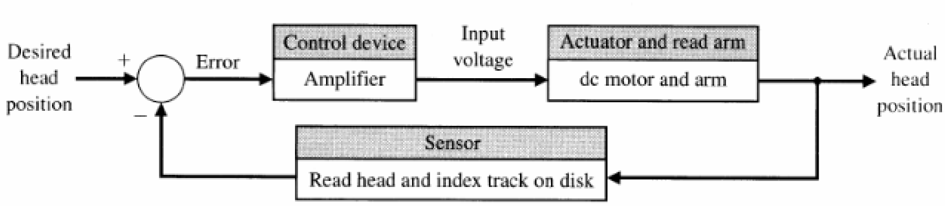


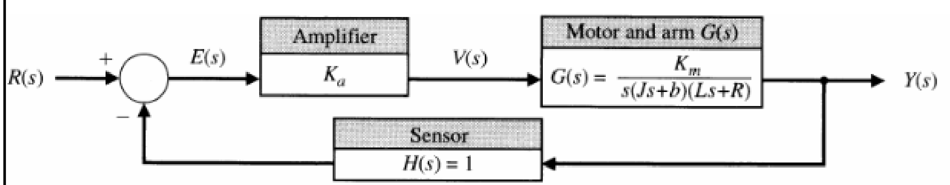





# Case Study-1: Disk drive

- **Block diagram**




Block diagram model of disk drive read system

# Case Study-1: Disk drive

- **Block diagram**

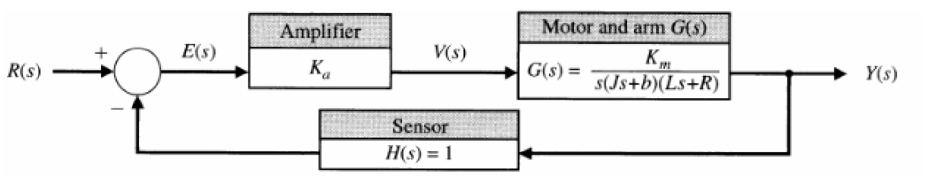
Typical parameters of Disk Drive Reader

| Parameter | Symbol | Typical value |
|---|---|---|
| Inertia of arm and read head | J | 1 N.m.s²/rad |
| Friction | b | 20 kg/m/s |
| Amplifier | $K_a$ | 10-100 |
| Armature resistance | R | 1Ω |
| Motor constant | $K_m$ | 5 N.m/a |
| Armature inductance | L | 1 mH |



# Case Study-1: Disk drive

- **Block diagram**



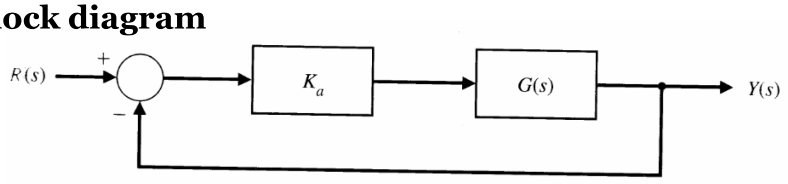

$$G(s) = \frac{K_m}{s(Js+b)(Ls+R)} = \frac{K_m / bR}{s(\tau_L s+1)(\tau s+1)}$$

Where $\tau_L$=J/b=50 ms and $\tau$=L/R=1 ms. Since $\tau << \tau_L$, we often neglect $\tau$. Then

$$G(s) = \frac{K_m / bR}{s(\tau_L s+1)} = \frac{0.25}{s(0.05s+1)} = \frac{5}{s(s+20)}$$

# Case Study-1: Disk drive

- **Block diagram**



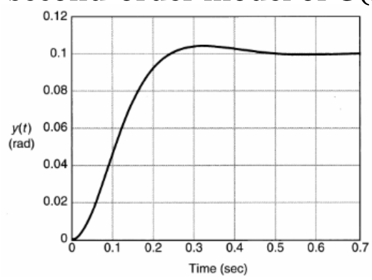

$$\frac{Y(s)}{R(s)} = \frac{K_a G(s)}{1 + K_a G(s)}$$

Using the approximate second-order model of G(s) we obtain:

$$\frac{Y(s)}{R(s)} = \frac{5K_a}{s^2 + 20s + 5K_a}$$

When $K_a$=40, we have

$$Y(s) = \frac{200}{s^2 + 20s + 200} R(s)$$




The system response for step input R=0.1/s rad



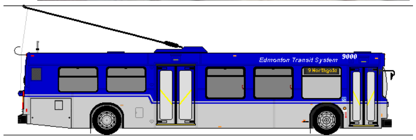
# Outline

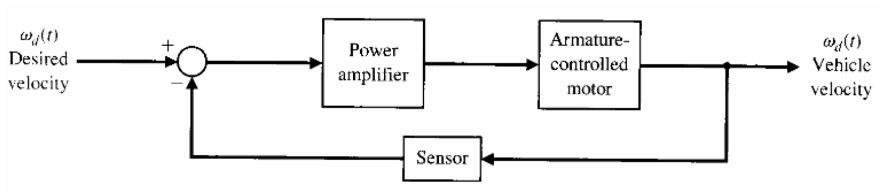
- Feedback Systems
- Feedback using Extrinsic Elements
- Feedback using Intrinsic Elements
- Case Study-1: Disk Drive
- **Case Study-2: Electric Traction Motor**
- Summary

# Case Study-2: Electric Traction Motor

A majority of modern trains and local transit vehicles utilize electric traction motors.

The electric motor drive for a railway vehicle incorporates the necessary control of the velocity of the vehicle.



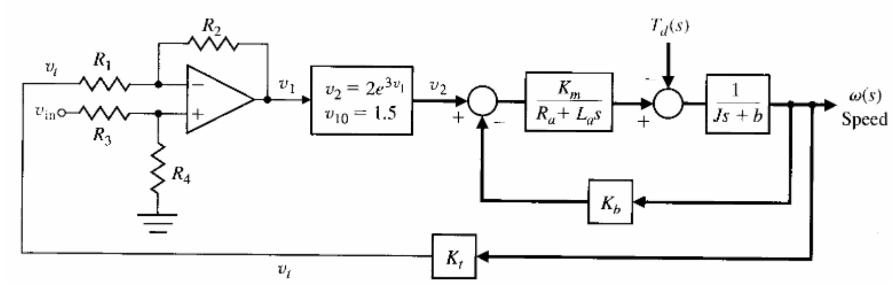





# Case Study-2: Electric Traction Motor

- **Modeling**

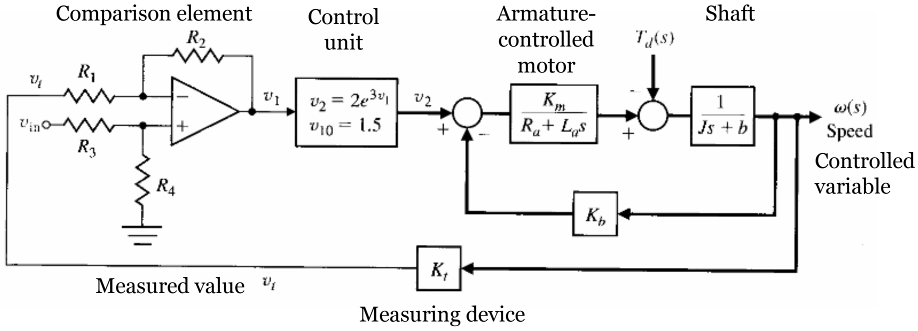
  The first step is to describe the transfer function of each block. We can use a tachometer to generate a voltage proportional to velocity and to connect that voltage, $v_t$, to one input of a difference amplifier.



# Case Study-2: Electric Traction Motor

- **Modeling**

  The power amplifier is nonlinear and can be approximately represented by $v_2 = 2e^{3v_1} = 2\exp(2v_1) = g(v_1)$ , an exponential function with a normal operating point, $v_{10}$=1.5v.






# Case Study-2: Electric Traction Motor

- **Modeling**

  Using the linearization technique:

$$v_2 = 2e^{3v_1} = 2\exp(2v_1) = g(v_1)$$

$$v_2 = \left[\frac{dg(v_1)}{dv_1}\right]\Delta v_1 = 2[3\exp(3v_{10})]\Delta v_1 = 2[270]\Delta v_1 = 540\Delta v_1$$

Then, discarding the delta notation and writing the Laplace transform, it follows that

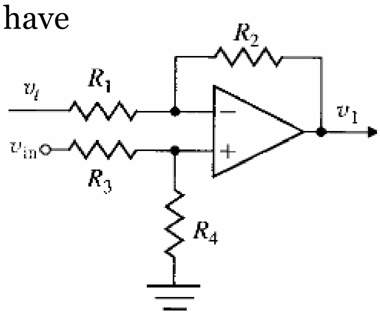
$$V_2(s) = 540V_1(s)$$

## Case Study-2: Electric Traction Motor

- **Modeling**

Also, for the differential amplifier, we have

$$v_1 = \frac{1 + R_2/R_1}{1 + R_3/R_4} v_{in} - \frac{R_2}{R_1} v_t$$




We wish to obtain an input control that sets $\omega_d(t)=v_{in}$ where the units of $\omega_d$ are rad/s and the units of $v_{in}$ are volts. Then, when $v_{in}$=10v, the steady-state speed is $\omega$=10 rad/s. We note that $v_t=K_t\ \omega_d$ is steady state.

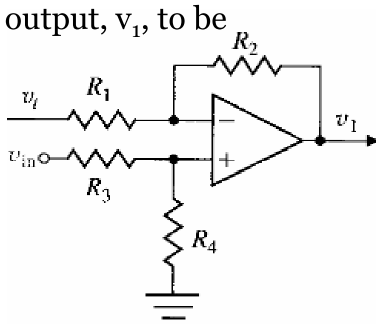


## Case Study-2: Electric Traction Motor

- **Modeling**

In balance, we expect the steady-state output, $v_1$, to be

$$v_1 = \frac{1 + R_2/R_1}{1 + R_3/R_4} v_{in} - \frac{R_2}{R_1} K_t(v_{in})$$




When the system is in balance, $v_1$=0, and $K_t$=0.1, we have

$$\frac{1 + R_2/R_1}{1 + R_3/R_4} = \left(\frac{R_2}{R_1}\right) K_t = 1$$

This relation can be achieved when

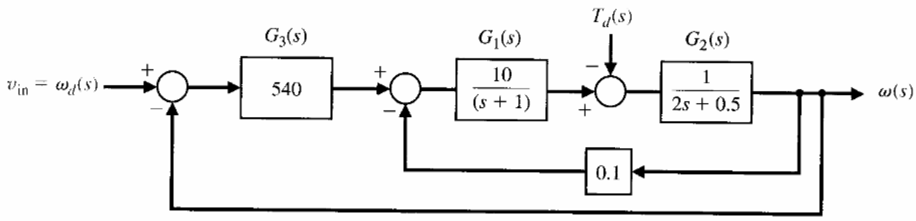
$$R_2/R_1 = 10 \quad \text{and} \quad R_3/R_4 = 10$$

# Case Study-2: Electric Traction Motor

- **Modeling**

Parameters of a large dc Motor

| | | | |
|---|---|---|---|
| J | 2 N.m.s²/rad | $R_a$ | 1Ω |
| b | 0.5 kg/m/s | $K_m$ | 10 N.m/a |
| $K_b$ | 0.1 | L | 1 mH |




$$\frac{\omega(s)}{\omega_d(s)} = \frac{540G_1(s)G_2(s)}{1+0.1G_1G_2+540G_1G_2} = \frac{540G_1(s)G_2(s)}{1+540.1G_1G_2} = \frac{5400}{(s+1)(2s+0.5)+5401}$$

$$= \frac{5400}{2s^2+2.5s+5401.5} = \frac{2700}{s^2+1.25s+2700.75}$$

Assuming that $T_d$=0



# Outline

- Feedback Systems
- Feedback using Extrinsic Elements
- Feedback using Intrinsic Elements
- Case Study-1: Disk Drive
- Case Study-2: Electric Traction Motor
- **Summary**

# Summary

- Ideal physical model can be obtained by schematically decomposing the real physical system into ideal building blocks; composed of resistors, masses, beams, kilns, isotropic media, Newtonian fluids, electrons, and so on.
- An ideal mathematical model can be obtained by applying natural laws to the ideal physical model; composed of nonlinear partial differential equations, and so on.
- Reduced mathematical model can be obtained from the ideal mathematical model by linearization, lumping, and so on; usually a rational transfer function.



# Summary

- No mathematical system can precisely model a real physical system; there is always uncertainty. Uncertainty means that we cannot predict exactly what the output of a real physical system will be even if we know the input, so we are uncertain about the system. Uncertainty arises from two sources: unknown or unpredictable inputs (disturbance, noise, etc.) and unpredictable dynamics.

# *Questions?*